Can Sovereign Wealth Fund Disclosures Be Monetized?
A Disclosure-Lag-Aware Study of PIF and Norges Bank Signals

Ethan Wei
GAO Capital
Singapore

Abstract—This paper examines whether public sovereign wealth fund (SWF) disclosures contain investable information once disclosure lag, partial visibility, and implementation constraints are treated as first-class design variables. The project began with three funds—Saudi Arabia's Public Investment Fund (PIF), Norges Bank Investment Management (NBIM), and Singapore's GIC—but quickly encountered radical disclosure asymmetry: NBIM provides broad but slow public-equity visibility, PIF provides security-level visibility only for its reportable U.S. 13F sleeve, and GIC is too opaque to support a comparable holdings panel. We therefore construct fund-specific normalized datasets and test lag-aware strategies rather than forcing all funds into a common template.

For PIF, five backtests are implemented on a canonical 13F panel with filing-aware trade timing, benchmark comparison against SPY, and a corrected split-adjusted price layer. After validation, all strategies underperform on an excess-return basis; the best cash-aware variant still trails the benchmark materially. For NBIM, eight strategies are tested against VT using split-adjusted daily prices for first-post-release execution and month-end marks derived from the same daily source. The strongest direct mirror results are shown to be contaminated by survivor-selection in a bounded mega-cap universe. More credible sector-based tests produce only modest benchmark-relative gains, with the best results coming from industry weight-change and concentrated leadership strategies. A final robustness layer then stresses the surviving signals with execution lags, explicit one-way costs, realistic caps, subperiod splits, and alternate benchmarks. An attribution layer decomposes benchmark-relative performance into exposure timing, allocation, and concentration effects. Across both funds, the evidence does not support strong claims of easy copy-trading alpha. The most credible residual signal is slower-moving sector posture rather than single-name imitation, and among the tested strategies only NBIM's concentrated leadership sleeve remains meaningfully positive through the full robustness stack.

Index Terms—sovereign wealth funds, 13F, NBIM, PIF, disclosure lag, backtesting, sector rotation, alpha

I. Introduction

Large sovereign wealth funds (SWFs) are unusual public investors. They are patient, structurally large, politically salient, and often treated by market participants as "strong hands" rather than tactical traders. This creates an intuitive hypothesis: if SWFs are informed, patient allocators, perhaps public disclosure of their holdings or reallocations contains tradable information.

That simple hypothesis becomes difficult once one confronts the actual disclosure regimes. Some funds disclose broad holdings but at long lag; others disclose only narrow regulatory subsets; others remain too opaque to support a serious panel. This project was designed around that asymmetry rather than around an idealized "all funds disclose the same thing" assumption.

The original target funds were:

- NBIM, manager of Norway's Government Pension Fund Global;
- PIF, Saudi Arabia's Public Investment Fund; and
- GIC, Singapore's sovereign wealth fund.

The original research questions were:

1) Can an investor harvest alpha by mirroring disclosed SWF positions after the information becomes public?
2) If single-name mirroring fails, can alpha still be extracted from the sectors or industries into which SWFs are rotating?
3) Are observable exits or reductions structured enough to improve timing or reduce stale-trade risk?

The project ultimately contributes four things:

1) a disclosure-aware research design for heterogeneous SWF datasets;
2) normalized holdings, transition, and benchmark layers for PIF and NBIM;
3) validated lag-aware backtests distinguishing exploratory from credible results; and
4) reproducible reports, charts, and interactive artifacts that expose the full path from disclosure to simulated trade.

II. Research Scope and Disclosure Regimes

A. Core Principle

The governing design rule was:

> A signal may only be traded after it is public, and no backtest may use a portfolio effective date as a trade date unless it is also the public date.

TABLE I
Disclosure Regimes Across Target Funds

Fund	Primary Public Source	Research Implication
NBIM	Public holdings database plus annual / half-year reports	Broad public-equity coverage, but low timeliness; suitable for slow-moving portfolio and industry tests.
PIF	SEC 13F filings	Security-level visibility with filing dates, share counts, and values; strongest candidate for lagged mirroring tests.
GIC	Annual reports and sparse external ownership filings	Too opaque for a comparable holdings panel; retained as a qualitative disclosure-asymmetry case.

TABLE II
Dataset Scale After Cleaning

Fund	Holdings Rows	Periods	Additional Scale Information
PIF	775	30	30 canonical filings, 6 amendments in raw set, 998 transition rows, 88 priced securities.
NBIM	97,010	11	98,450 transition rows, 16,445 unique issuer-country keys, 22 priced instruments with full daily coverage.

This rule is the main protection against overstated alpha in disclosure-based research.

B. Disclosure Regimes

Table I summarizes the three target funds.

The final quantitative panel therefore excludes GIC, not because the fund is unimportant, but because the public data was too thin to support a comparable holdings backtest without false precision [4], [5].

III. Data Construction Pipeline

A. Data Layers

The project was built around reusable data layers rather than one-off spreadsheets:

- Holdings panels: normalized security or issuer snapshots.
- Transition panels: entries, exits, and change classifications between snapshots.
- Summary panels: sector, geography, and concentration summaries by period.
- Backtest outputs: signal eligibility, rebalance events, order blotters, position histories, portfolio paths, and benchmark-relative summaries.

B. Dataset Scale

The final project scale is summarized in Table II.

C. PIF Data Pipeline

PIF data was pulled from EDGAR 13F filing folders containing primary_doc.xml and informationtable.xml, consistent with the SEC's 13F disclosure framework [1]. The pipeline:

1) parsed filing metadata and holdings rows from XML,
2) indexed amendments and canonical filing versions,
3) normalized security keys using CUSIP, class, and option flags,
4) built a transition dataset based on share-count and value changes, and
5) mapped approved securities to a validated price layer.

The transition dataset classified entry_observed, exit_observed, likely_accumulation, likely_reduction, and continued_holding. Because 13F includes share counts, these classifications could be anchored in actual share changes rather than inferred ownership percentages.

D. NBIM Data Pipeline

NBIM data was assembled from downloaded public equity files labeled by snapshot date (e.g., eq_YYYYMMDD) using NBIM's public holdings distribution [2], [3]. The pipeline:

1) stacked semicolon-delimited holdings files into a consolidated issuer-level panel,
2) normalized numeric ownership and voting fields,
3) generated annual and half-year summary datasets,
4) classified issuer transitions between snapshots, and
5) mapped industry exposures to liquid U.S. sector ETF proxies.

Unlike PIF, NBIM holdings did not include share counts. This matters. A change in market value alone does not prove buying or selling. Therefore, the NBIM transition engine treated inclusion/exclusion and ownership-percentage changes as the strongest available indicators and did not infer flows directly from value movement.

E. GIC Treatment

GIC remained in scope conceptually but out of scope for the quantitative backtests. The project documented the relevant proxy channels—major-shareholder filings, 13D/G, HKEX DI, FCA TR-1, EDINET, and deal press—but did not construct a comparable holdings panel because the public evidence was episodic, threshold-driven, and structurally incomplete.

IV. Price Layers, Benchmarks, and Trade Timing

A. Trade Timing

All strategies obeyed disclosure-lag-aware execution rules:

- PIF signal date: SEC public filing date.
- PIF trade date: next trading day after the public date.
- NBIM signal date: official annual or half-year publication date.

- NBIM trade date: first tradable close strictly after publication.

For NBIM, the portfolio is rebalanced at the first post-release daily close and then carried at month-end closes until the next report. This preserves realistic post-disclosure reaction timing without requiring full daily path simulation between reports.

B. Benchmarks

The benchmark selection matched the disclosed universe:

- PIF benchmark: SPY, reflecting a U.S.-listed 13F sleeve.
- NBIM benchmark: VT, reflecting a broad global public-equity owner.

C. Price Sources

Price infrastructure was fund-specific:

- PIF: Twelve Data daily price history, with both raw execution prices and an adjusted mark-to-market layer [7].
- NBIM: Twelve Data adjusted daily history for the benchmark, ETF proxies, and the bounded direct mirror universe; month-end marks are derived from this daily source [7].

This unified daily source removed the earlier timing distortion from waiting until the next month-end to enter an NBIM trade and also eliminated the earlier ETF proxy gaps.

V. Strategy Definitions

A. PIF Strategies

Five PIF strategies were tested:

1) P1 New Positions Mirror: buy new disclosed entries only.
2) P2 Full Sleeve Equal Weight: hold the full common-equity 13F sleeve equally weighted.
3) P3 Accumulation Tilt: overweight likely accumulations.
4) P4 Exit Avoidance: avoid likely reductions and exits.
5) P5 Cash-Aware Copy: copy buys and sells as disclosed while retaining sale proceeds in cash.

P5 was introduced after an important conceptual observation: if PIF liquidates positions from its disclosed sleeve, a literal copycat should often keep the residual cash rather than mechanically concentrating the survivors.

B. NBIM Strategies

NBIM strategies naturally fall into two families.

1) Exploratory direct mirrors:

1) N1 Core U.S. Mirror Equal Weight
2) N2 Core U.S. Mirror NBIM Weight

These were included because single-name mirroring was one of the original hypotheses, but they rely on a bounded U.S. mirror sleeve rather than a full global book.

2) Industry-based strategies:

1) N3 Industry Weight Mirror: reproduce the disclosed industry posture through ETF proxies.
2) N4 Industry Weight-Change Tilt: hold only sectors whose disclosed weights increased versus the prior snapshot.
3) N5 Industry Accumulation Tilt: hold sectors with positive accumulation-minus-reduction transition scores.
4) N6 Top-3 Industry Leaders: concentrate in the three largest disclosed sector exposures.
5) N7 Top-3 Industry Increases: hold the three largest positive sector changes.
6) N8 Consensus Rotation Tilt: require both positive weight change and positive transition score.

This second family is more realistic and more aligned with what NBIM actually discloses: slow-moving sector posture rather than actionable stock-level trading instructions.

C. Benchmark-Relative Measure

For clarity, the paper reports both standalone total return and benchmark-relative excess return. The latter is defined as

$$R_{s,T}^{\text{excess}} = \frac{\text{NAV}_{s,T}/\text{NAV}_{s,0}}{\text{NAV}_{b,T}/\text{NAV}_{b,0}} - 1, \tag{1}$$

where s is the strategy and b is the benchmark. This choice makes the interpretation straightforward: positive values represent outperformance over the matched benchmark horizon.

VI. Validation Protocol

A. Why Validation Changed the Interpretation

The project produced two major validation lessons.

1) PIF split handling: Early PIF results appeared dramatically positive. Investigation showed that raw close series were being marked against unchanged share counts through reverse splits, creating fake gains. Once the backtests were rebuilt on adjusted mark-to-market pricing, the apparent alpha disappeared and all fully invested PIF strategies became negative in benchmark-relative terms.

2) NBIM mirror-universe bias: The initial NBIM direct mirror results were also strikingly positive, but the issue was different. Here the arithmetic was fine; the interpretation was not. The bounded mirror universe turned out to be almost the same recurring basket of U.S. mega-cap winners in every snapshot. A mirror strategy built on that sleeve is closer to a hand-selected winners basket than to a neutral test of disclosure alpha.

B. Mechanical Integrity Checks

The final backtests were checked for:

- negative cash breaches,
- impossible gross exposure,
- weight-integrity gaps, and
- rebalance arithmetic consistency.

For the final NBIM panel, all strategies showed zero negative-cash breaches and only floating-point-level weight gaps. The corrected PIF engines passed the same integrity standard.

In addition, cached market-data rows were spot-checked against live API responses at representative benchmark, ETF, and single-name points after the final rebuild. The sampled VT, XLK, XLC, TSLA, UBER, and LCID closes matched the stored price layers exactly. A dedicated Phase-3 robustness audit then re-checked all A1–A3 outputs through 26 automated tests spanning baseline reproduction, delayed execution pricing, self-financing cost application, cap compliance at rebalance, and benchmark alignment; the final audit passed with zero failed checks.

VII. Results: PIF

A. Summary Results

Table III reports the validated PIF strategy outcomes. The important point is not just that most strategies fail, but that the best conceptual fix (cash-aware copying) still fails to generate alpha versus the benchmark.

B. Interpretation

Three findings are robust:

1) Naive mirroring fails. Security-level lagged copying of the disclosed 13F sleeve is not a strong stock-picking edge.
2) Cash-awareness helps but does not rescue alpha. The difference between P2–P4 and P5 shows that disclosed sales contain useful exposure information, but not enough to beat passive U.S. beta.
3) Partial disclosure is a structural limit. Even a mechanically faithful mirror of PIF's 13F book is not a mirror of the entire fund.

The implication is that PIF's public sleeve is better interpreted as a monitoring and posture dataset than as a direct alpha source.

VIII. Results: NBIM

A. Exploratory Mirrors versus Realistic Signals

Table IV reports the final NBIM strategy results versus VT. Unlike the PIF panel, the table must be interpreted in two layers: the exploratory direct mirrors and the realistic sector-based tests.

B. Why N1 and N2 Are Not Strong Alpha Evidence

The most important NBIM result is not the high return of N1 and N2, but the reason that those returns cannot be treated as robust disclosure alpha. The mirror universe was a bounded U.S. mega-cap basket, and the bias audit showed that it overlapped almost perfectly with the top recurring U.S. names in the NBIM book across the sample. In other words, the backtest was effectively long a persistent basket of secular winners.

That is a valid exploratory result—it tells us what a bounded, disclosure-informed mega-cap sleeve would have done—but it is not clean evidence that the public disclosure stream itself creates a broadly harvestable copy-trading edge.

C. What Looks More Credible

The realistic NBIM findings are much narrower:

- N3 is essentially benchmark-like. Broadly mirroring NBIM's sector posture mostly reproduces a high-quality global allocation.
- N4 is the strongest realistic signal. Following sectors whose disclosed weights increased produced a modest positive excess return of 18.6% versus VT.
- N6 also worked reasonably well. Concentrating on NBIM's top three sector exposures produced an 18.1% excess return.
- N5, N7, and N8 are weak or too sparse. Transition-only and consensus-only rules do not appear strong enough on their own in this implementation.

Thus, the most credible NBIM signal is not direct mirroring but slow-moving sector posture, especially disclosed industry weight change.

IX. Results: Combined PIF + NBIM Strategies

A. Phase-2 Combined Results

The Phase-2 tests ask a narrower and more realistic question than either single-fund panel alone: can PIF's exposure-regime information and NBIM's sector-posture information be combined into a modest but credible overlay strategy? Table V summarizes the results.

Three observations stand out.

1) S1 is the only positive VT-relative result. The combination of a PIF risk regime and an NBIM sector overlay can add a small amount of value versus VT, but the edge is modest.
2) Consensus-only confirmation is too restrictive. S2 spends too much time in cash or fallback exposure, which limits both upside and signal harvest.
3) PIF helps more as a throttle than as a selection engine. S4 and S5 are cleaner than S3 because they preserve validated NBIM sleeves and only use PIF to modulate gross exposure. Even so, they still fail to beat VT.

The combined evidence therefore strengthens, rather than weakens, the conservative interpretation of the project. There is some informational value in the interaction between the two funds, but it appears to be a risk-and-sector overlay effect rather than a strong source of standalone alpha.

B. Phase-3 Robustness Results

The next question is whether any surviving signals remain after more realistic implementation stress. Four focus strategies were selected for robustness work:

- P5: the best PIF design conceptually, because it retains cash after disclosed sales;
- N4: the strongest NBIM weight-change strategy;

TABLE III
PIF Strategy Results versus SPY

Strategy	Total Return	CAGR	Max Drawdown	Excess Return vs SPY	Information Ratio
P1 New Positions Mirror	−75.8%	−20.2%	−94.8%	−90.5%	−0.299
P2 Full Sleeve Equal Weight	−32.3%	−5.2%	−81.1%	−77.4%	−0.319
P3 Accumulation Tilt	−69.3%	−15.0%	−88.7%	−89.8%	−0.415
P4 Exit Avoidance	−45.2%	−7.9%	−80.9%	−81.7%	−0.362
P5 Cash-Aware Copy	+48.9%	+5.6%	−62.6%	−50.4%	−0.120

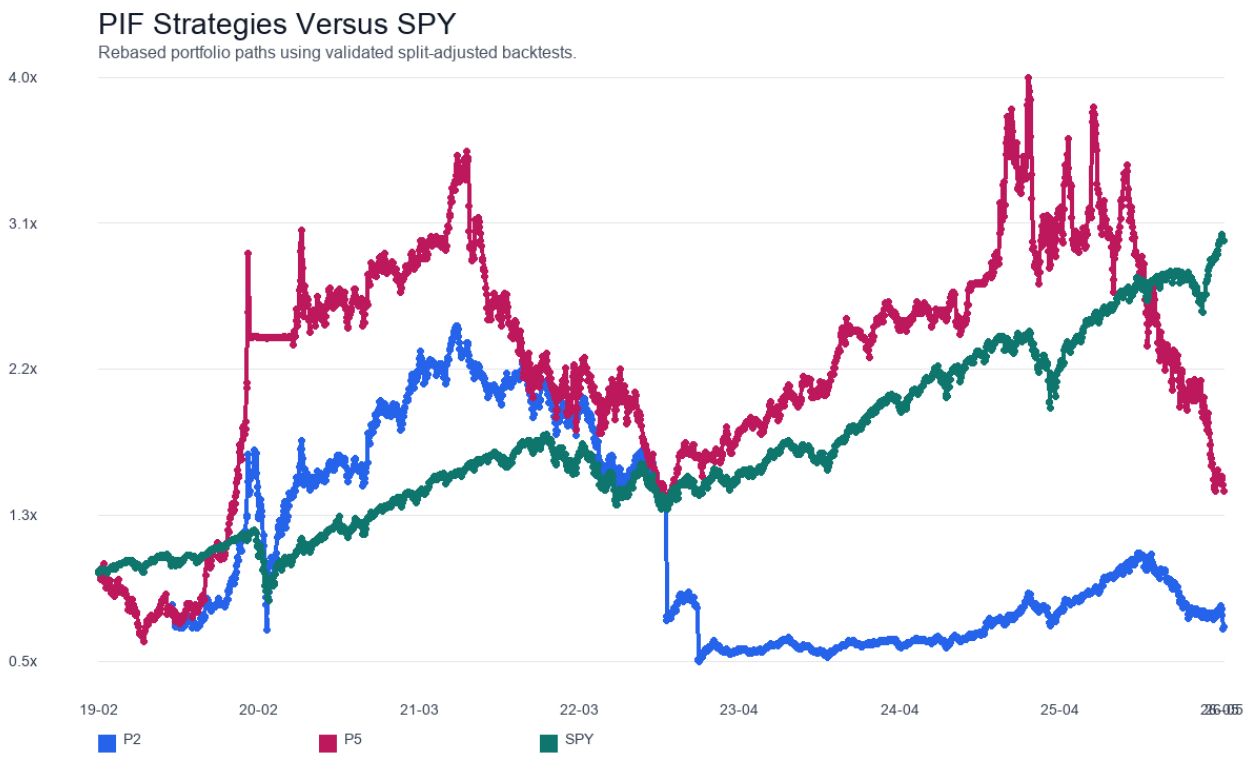

Fig. 1. Validated PIF strategy paths versus SPY. The cash-aware strategy (P5) materially improves on fully invested variants but still lags the benchmark.

TABLE IV
NBIM Strategy Results versus VT

Strategy	Type	Total Return	Ann. Return	Max Drawdown	Excess Return vs VT
N1 Core U.S. Mirror Equal Weight	Exploratory mirror	+717.0%	33.6%	−28.6%	+231.7%
N2 Core U.S. Mirror NBIM Weight	Exploratory mirror	+466.8%	27.0%	−34.7%	+119.8%
N3 Industry Weight Mirror	Broad industry mirror	+161.1%	14.2%	−22.2%	+2.1%
N4 Industry Weight-Change Tilt	Targeted rotation	+199.0%	16.3%	−24.3%	+18.6%
N5 Industry Accumulation Tilt	Targeted transition tilt	+69.2%	7.5%	−24.3%	−32.9%
N6 Top-3 Industry Leaders	Targeted concentration	+197.7%	16.2%	−25.8%	+18.1%
N7 Top-3 Industry Increases	Targeted rotation	+129.9%	12.2%	−24.4%	−8.8%
N8 Consensus Rotation Tilt	Targeted consensus filter	+83.3%	8.7%	−19.2%	−27.3%

- N6: the strongest concentrated NBIM leadership sleeve; and
- S1: the best combined PIF + NBIM overlay.

Three robustness dimensions were tested:

1) A1 execution lag: baseline next-day trading versus T+3 and T+5;
2) A2 explicit one-way execution costs: 0, 10, 25, and 50 bps;
3) A3 concentration and exposure caps: realistic position, sector, and gross-exposure limits.

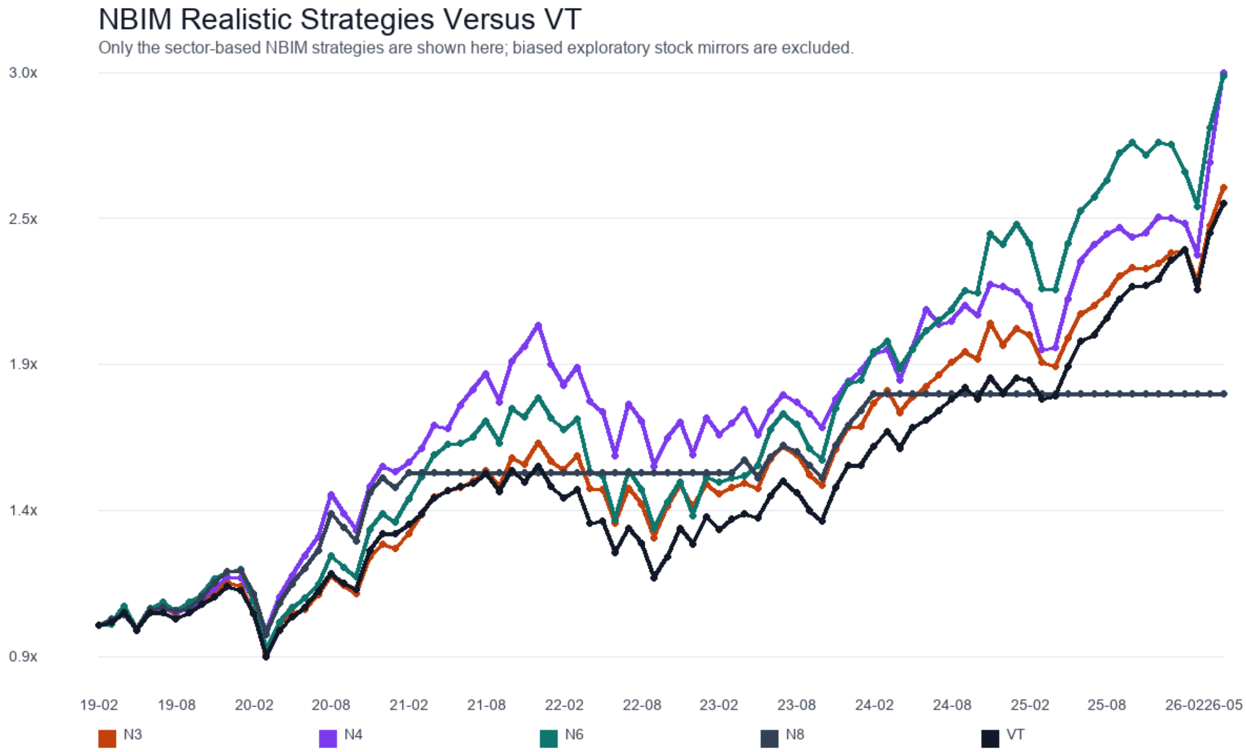

Fig. 2. NBIM sector-based strategies rebased against VT. Only the realistic industry strategies are shown; the exploratory direct mirror sleeve is intentionally excluded from this figure.

TABLE V
Combined PIF + NBIM Strategy Results

Strategy	Total Return	Excess Return vs VT	Excess Return vs SPY	Max Drawdown	Avg. Cash Weight
S1 Exposure Regime Overlay	+154.7%	+1.9%	−15.1%	−25.1%	27.0%
S2 Cross-Fund Consensus Sector Tilt	+82.3%	−27.1%	−39.2%	−22.1%	42.4%
S3 PIF Cash-Aware Base + NBIM Overlay	+58.0%	−36.8%	−47.3%	−60.2%	7.8%
S4 N4 Sleeve With PIF Risk Filter	+134.4%	−6.2%	−21.9%	−24.3%	27.6%
S5 N6 Sleeve With PIF Risk Filter	+133.9%	−6.4%	−22.0%	−29.4%	27.2%

Table VI compresses the most decision-relevant results. It reports the baseline benchmark-relative excess return together with the same measure under the most demanding lag, cost, and cap variants tested.

The interpretation is sharper than the Phase-2 results alone:

- P5 remains non-alpha. The strategy is still useful for understanding PIF as an exposure-throttle signal, but it does not beat SPY under any tested implementation variant.
- N4 is plausible but cap-sensitive. Its positive excess return survives delayed execution and one-way costs, but disappears once realistic sector caps are imposed.
- N6 is the strongest residual signal. It remains positive versus VT under delayed execution, explicit costs, and both moderate and tight cap regimes.
- S1 is fragile. The modest Phase-2 edge vanishes once execution is delayed or even modest costs and exposure constraints are imposed.

This robustness layer matters because it narrows the project's actionable conclusion. The surviving signal is not "copy SWFs" in general, nor even "combine PIF and NBIM." It is specifically that a simple, slow-moving NBIM sector-leadership sleeve appears to carry some residual information even after several realistic implementation stresses.

The subsequent subperiod-stability test reinforces the same hierarchy. P5 is strong only in the early sample and gives back the edge in 2024–2026. S1 is positive in 2019–2021 but negative in the later subperiods. N4 is regime-dependent: strong in 2019–2021, negative in 2022–2023, and only modestly positive thereafter. N6 is the only focus-set strategy that remains positive versus VT in all three subperiods, although its excess return becomes very small

TABLE VI
Phase-3 Robustness Summary for the Focus Set

Strategy	Benchmark	Baseline Excess	T+5 Excess	50 bps Excess	Tight-Cap Excess
P5 Cash-Aware Copy	SPY	−151.1%	−149.0%	−161.2%	−156.7%
N4 Industry Weight-Change Tilt	VT	+71.2%	+70.8%	+52.1%	−48.8%
N6 Top-3 Industry Leaders	VT	+64.4%	+62.9%	+60.0%	+33.7%
S1 Exposure Regime Overlay	VT	+4.7%	−8.7%	−14.3%	−79.0%

in the most recent window. This is not evidence of a large persistent alpha engine, but it is stronger support for N6 than for any other tested strategy in the project.

C. Benchmark Robustness and Attribution

Phase 3 then asked whether the remaining conclusions depend too heavily on benchmark choice. This matters especially for two cases: P5, where a U.S. 13F sleeve might look different against growth-heavy benchmarks, and S1, where a mixed cash-and-sector strategy might appear stronger against a global benchmark than against a U.S. benchmark.

The benchmark-robustness results sharpen the interpretation further:

- P5 remains non-alpha under both benchmark choices. It underperforms SPY by 151.1% and underperforms QQQ by 282.9% on the matched live window.
- N4 and N6 remain positive under an alternate global benchmark. N4 posts 71.2% excess versus VT and 69.4% excess versus ACWI. N6 posts 64.4% excess versus VT and 62.7% versus ACWI.
- S1 is benchmark-sensitive. It is slightly positive versus VT at 4.7%, but negative versus SPY at −45.4% and still negative versus a 50/50 rebased VT/SPY blend at −20.4%.

That result is important because it separates "global-allocation-like" behavior from true cross-market alpha. S1 appears closest to a modest global-overlay effect, not to a benchmark-agnostic source of excess return.

The attribution layer then explains why the four focus strategies behave differently. The decomposition is performed in exact daily arithmetic excess-return space, where each period's excess return is split into:

1) an exposure timing effect, driven by holding more or less benchmark exposure through cash;
2) an allocation effect, measured by equal-weighting the invested sleeve; and
3) a concentration effect, measured as the incremental impact of the actual strategy weights versus equal weights within that same sleeve.

The resulting interpretation is highly consistent with the earlier robustness findings:

- P5 fails mostly because of concentration, not because of cash alone. Its cumulative arithmetic cash drag is small at roughly −0.9 percentage points, while the concentration term is strongly negative at about −105.7 percentage points.
- N4 relies heavily on concentration. Its allocation term is negative, but a strongly positive concentration term overwhelms it. This fits the cap-sensitivity finding from A3.
- N6 is the cleanest survivor. Its allocation term is strongly positive, while concentration is near zero. That is exactly the pattern one would want from a robust sector-leadership sleeve.
- S1 gives back much of its potential edge through cash drag. It has a positive allocation term, but a large negative exposure-timing effect from running below full benchmark exposure.

This attribution result is arguably the most decision-useful output of Phase 3. It shows that N6 is not merely surviving by accident or by leverage-like concentration. Instead, its residual signal appears to come from the sectors it chooses to own rather than from aggressive weight concentration.

X. Discussion

A. What the Combined Evidence Says

The PIF and NBIM panels point to the same high-level conclusion from different directions:

- direct single-name copy trading is weak once lag and partial visibility are respected;
- broad mirroring often collapses into benchmark-like exposure rather than differentiated alpha; and
- the most plausible residual signal lies in slower-moving allocation, sector rotation, and risk throttling.

In PIF, even the most sensible copycat variant fails to beat SPY. In NBIM, the strongest realistic sector-based strategies are positive, but only modestly so. In the combined panel, only S1 exceeds VT, and only by a small amount; the filtered-sleeve variants S4 and S5 remain below both VT and SPY.

Phase 3 further tightens that conclusion. Once lag sensitivity, explicit costs, concentration constraints, subperiod stability, and alternate benchmark choices are modeled, the combined edge in S1 no longer looks robust. N4 remains interesting but cap-sensitive and concentration-dependent. N6 is the only strategy in the focus set that stays meaningfully positive through all tested robustness layers, and its attribution profile is cleaner than the alternatives.

B. What Was Built Beyond the Final Returns

This project produced more than terminal backtest numbers:

- normalized holdings and transition datasets for PIF and NBIM;
- audit layers for disclosure dates, price coverage, and mirror-universe bias;
- benchmark-relative reports and chart packs;
- a PIF interactive explorer for stepping through filings, signals, orders, and portfolio evolution; and
- a reusable structure for extending the project to future monitoring and higher-capacity pricing sources.

This matters because disclosure-based alpha is especially vulnerable to hidden assumptions. The system-level outputs were designed to make those assumptions inspectable.

XI. Limitations

Several limitations remain:

- market impact, taxes, and venue-specific implementation frictions were not modeled even though Phase 3 did add explicit lag, transaction-cost, cap, subperiod, and benchmark sensitivity;
- GIC was not included quantitatively because the public data was too incomplete;
- NBIM sector tests still used U.S. ETF proxies for a global industry book, introducing representation error;
- the NBIM direct mirror universe remained bounded and should still be treated as exploratory even after the timing and daily-price corrections;
- the strongest NBIM direct mirror results were contaminated by mirror-universe selection bias;
- the attribution layer is exact in arithmetic excess-return space, but it should not be confused with a perfect decomposition of compounded terminal return.

These do not invalidate the project, but they sharply limit how strong any alpha claim can be.

XII. Conclusion

This study asked whether sovereign wealth fund disclosures can be monetized after realistic lag and visibility constraints are enforced. The answer is mostly negative for naive copy trading and only cautiously positive for narrow sector-based signals.

For PIF, no strategy generated alpha versus SPY. The best-performing design, cash-aware copying, improved materially on fully invested mirroring but still lagged passive beta.

For NBIM, the strongest raw stock-mirror result could not be accepted at face value because it was driven by a bounded basket of recurring mega-cap winners. More realistic industry-based strategies were far more credible, but the edge was modest. The best signals came from industry weight changes and concentrated leadership sectors, not from broad portfolio mirroring.

For the combined Phase-2 layer, the conclusion is similar. Using PIF as a gross-exposure regime and NBIM as a sector allocator can produce a small benchmark-relative edge versus VT, but the effect is not large enough to support a strong alpha claim and does not survive the tougher implementation tests in Phase 3 or the broader benchmark comparisons.

The highest-confidence conclusion is therefore:

> Public SWF disclosures are more useful as slow-moving allocation and rotation signals than as literal copy-trading instructions, and even then the most durable evidence in this project narrows to a modest NBIM sector-leadership effect rather than a broad cross-fund alpha engine.

The practical continuation of this research should prioritize event-level attribution, state-transition analysis, and monitoring tools, rather than ever-larger naive mirroring engines.

References

[1] U.S. Securities and Exchange Commission, "Frequently Asked Questions About Form 13F," https://www.sec.gov/rules-regulations/staff-guidance/division-investment-management-frequently-asked-questions/frequently-asked-questions-about-form-13f.
[2] Norges Bank Investment Management, "Investments / All Investments," https://www.nbim.no/en/investments/all-investments/.
[3] Norges Bank Investment Management, "Terms of Use – Holdings Data in Snowflake Marketplace," https://www.nbim.no/en/terms-of-use-holdings-data-in-snowflake-marketplace/.
[4] GIC, "Reports," https://www.gic.com.sg/newsroom/reports/.
[5] GIC, "Santiago Principles / Institutional Framework and Governance Structure," https://www.gic.com.sg/how-we-invest/santiago-principles/institutional-framework-and-governance-structure/.
[6] Alpha Vantage, "Official Documentation," https://www.alphavantage.co/documentation/.
[7] Twelve Data, "API Documentation," https://twelvedata.com/docs.